

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing FINAL YEAR PROJECT PROPOSAL & PLAN

Edu Path

Project Team

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[Project Supervisor]
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Edu Path

Change Record

Author(s)	Version	Date	Notes	Supervisor's Signature
	1.0		Original Draft	
			Changes Based on Feedback From Supervisor	
			Changes Based on Feedback From Faculty	
			Added Project Plan	
			Changes Based on Feedback From Supervisor	

Project Proposal

Project Title: AI powered Academic and Decision Support System (Edu Path)

Opportunity & Stakeholders:

Edu Path is a chance to build an AI-powered system that gives advice to students who want to go to University. The system will use technology to suggest areas of study, degrees and schools based on what students have studied before what they like, what they are good, at what they want to do in the future how much money they have and where they live. The system will make it easier for students to find the choices that match their needs and what they like. The main people who will use Edu Path are students who get help from the advice the system gives. Other people who are involved are parents or the people who take care of the students.

Existing System/ Description of the Current Situation:

Currently, several online platforms are available to help students explore higher-education opportunities. Platforms such as EduPath Career Pathway provide career and educational pathway guidance, while [educations.com](https://www.educations.com) offer program search, university information, scholarships, study destinations, and program-matching features. Similarly, [GoToUniversity.com](https://www.gotouniversity.com) provides university and course search with different filters and information such as tuition, rankings, and university details. These existing systems make educational information more accessible to students, but students may still need to make different decisions separately regarding their suitable academic field, degree program, and university. Edu Path differs from these existing systems by providing an integrated and personalized recommendation process. Instead of only searching or matching available programs and universities, Edu Path first analyzes the student's complete profile, including academic background, interests, skills, career goals, budget, location, and preferences, to recommend a suitable academic field. After that, it recommends relevant degree programs, and finally filter and rank suitable universities according to the

student's individual requirements. Therefore, Edu Path connects the complete decision-making process from Academic Field → Degree Program → University within a single personalized system.

Problem Statement:

Students getting intermediate education often struggle to make a choice concerning which academic field, degree program, and university is suitable for them. The decision is hard since students differ in academic background, interests, skills, career objectives, financial status, and location preference. Currently, students usually gather information from numerous university websites, educational platforms, counselors, and other sources. The process is tedious since useful information is scattered over different resources requiring a lot of effort to review and select what suits one's purpose. The existing platforms provide university and program information search engines, search facilities, and in some cases program or career matching. However, students have to make separate decisions regarding their academic field, program, and university. Most platforms do not offer a personalized process of matching academic fields, the program, and the universities. Students are also expected to consider several aspects including eligibility, costs, scholarships, location, and university preferences when choosing a university. This makes the process of comparing universities according to these criteria very challenging. Therefore, there is a need for an integrated system that understands the student's global academic needs. A system that provides academic field recommendations, relevant degree programs, and suggested universities based on specific criteria would be more efficient for students seeking higher education.

Proposed Solution:

Edu Path is envisioned to be an innovative AI-assisted personalized academic and university recommendation system for students having completed intermediate education and looking to pursue higher education. The system will be designed to acquire the academic background, interests, and skills, career, budget, location, and university preferences of the student through a web-based system. The system will utilize the information to form a student profile upon which a Machine Learning model will

recommend an academic field in higher education. Once an academic field has been determined, the system will use LLM with Retrieval-Augmented Generation to find relevant degree programs in higher education from university and program knowledge base and match the programs with the student's interests and career goals. The system will then apply eligibility and preference-based filtering using factors such as program availability, budget, location, scholarships, and university preferences. The system will finally use MCDM/TOPSIS to rank the eligible universities based on the student selected criteria and priorities. The system will present the student with a personalized recommendation consisting of a suitable academic field, relevant degree programs, and ranked universities based on the recommendations with supporting information for the recommendations. University and program information will be sourced from existing datasets and university sources, and web crawling will be used to update the available information periodically.

Scope of the Project:

- **Student Profile**

This module will gather and collate the student's academic records, interests, skills, career, budget, location, and university preferences to create an organized student profile. This will act as the primary input to all the recommendation modules. The information will be utilized to understand individual students' requirements.

- **Voice + Text NLP**

The module will allow the students to enter their interests, goals, and preferences. Students can input their requirements using natural language (text and/or voice). The speech will be converted to text in the case of voice input. NLP and LLM techniques will be used to extract required information from the natural language query. This will contribute to the student's profile.

- **Eligibility Check**

This module will evaluate if a student is eligible for the selected academic program or university. The student's academic profile will be compared to the eligibility criteria

(relevant subjects, grades, etc.) of the selected programs or universities. This will highlight eligible and ineligible options for each student. This helps in avoiding recommendations of programs or universities for which one does not meet the minimum requirements. The results will also be used for university filtering.

- **Personalized Program Recommendation**

This module will recommend degree programs that suit the academic background, interests, skills, and career of the student. Relevant program descriptions will be retrieved using RAG from the university and program knowledge bases. Semantic similarity techniques will be used to identify relevant programs based on the student's interests and career. An LLM will interpret the retrieved information and link the information to the student's requirements. The module will be personalized to avoid giving generic program lists.

- **University Ranking**

This module will rank the universities that offer the recommended programs and satisfy the eligibility requirements. The ranking will be done against several factors: academic compatibility, interest and career fit, budget, location, scholarships, and other preferences. MCDM/TOPSIS methods will be used to make the selection of the best university for a student based on the specified criteria. The weightage of criteria could be according to individual students' preferences. As such, two students with different preferences will get different university rankings.

- **University Comparison**

The module will enable students to compare shortlisted universities according to relevant factors. Factors such as program fees, eligibility, program availability, location, scholarships, and program or career fit will be compared. This will help the student in making the final decision based on their requirements. The comparison will rely on the information available in the university knowledge base. The comparison will not make the admission decisions on a student's behalf.

- **Women-Only University Recommendation**

This module will offer gender-based university recommendations (wherever applicable). If a student selects a female-only university preference, the system will filter university options. This module will be used in conjunction with program availability, eligibility, location, and other recommendation criteria. It will ensure that the recommendations meet the selected preference. The feature will be available wherever gender-based university information is present.

- **Skill Gap Analysis**

This module will highlight skill gaps based on one's career or academic path. It will identify skills that one already has and those that one lacks or needs to develop. The information will assist the student in developing their career path and academic skills. The module will rely on the available career and program skill gap information.

- **Adaptive/Dynamic Questions**

This module will ask dynamic questions depending on the student's input and profile. The same question will not be asked to all students; rather, the system will identify relevant questions based on the student's interests and requirements. For instance, a student interested in the computing field will be asked computing-related questions. A student concerned with budgets and scholarships will be asked related questions. Additionally, RAG can be used to retrieve relevant verified information to respond to student questions whenever appropriate. This will make the information collection more student-centric.

- **Explainable Recommendations**

This module will explain the rationale behind each recommendation. It will explain why a certain recommendation (academic field, degree program, and university) was made based on the criteria selected by the student and their profile. For the case of ranking universities, the factors considered in the ranking (budget, location, eligibility, etc.) will be shared so that the student can be informed. It will not guarantee any specific result but will explain how the recommendation was made.

- **Data Freshness**

This module will keep track of the source and the date of data verification of the information used by the system. Applicable information will be collected from existing datasets and publicly accessible university information sources. Web crawling can be used to collect updated information at regular intervals (if the information is publicly available). The information's authenticity and freshness will be identified by the date on which it was last verified. It will not be guaranteed that the information is up-to-date since it will depend on the frequency at which new data is crawled and verified.

List of Faculty Proposed Changes

Project Title

Supervisor's Signature: _____

Proposed Change	Proposed By	Supervisor's Decision
The student has limited background knowledge of the proposed domain. Moreover, the problem statement is not clearly defined. The student should strengthen their domain knowledge and refine the problem statement before proceeding with the project. Work on suggestions provided. Perform some more literature review to achieve the objective.	Sir Mubariz Rehman	Approved
Work on suggestions provided. Perform some more literature review to achieve the objective.	Sir Usman Sharif	Approved
Proposed Solution is not well defined.	Sir Usman Kareem	Approved
Students need to conduct market survey. They only conducted research paper based surveys which is insufficient to produce a product that will justify efforts to build FYP. Accepted with condition that students will do market survey to find existing applications.	Sir Tabbassum Javed	Approved
Lot of Lackness in literature, do it more to get clarity how you will achieve your target. Work on suggestions.	Sir Tajamul Shehzad	Approved
Needs better proposed solution, and major revision and research for methodology.	Sir Hammad Ahmed	Approved
Review existing academic and career decision-support systems to better understand their working, functionality, scope, and limitations.	Sir Zeeshan Ali	Approved
Suggestions: The scope of the project is required to be well-defined. Currently, it is too open. Comments: The students committed to develop a smartphone application + Web portal.	Sir Sajid Qureshi	Approved

Project Plan

Work Breakdown Structure: A work breakdown structure (WBS) is deliverable based decomposition of project scope. The WBS includes 100% of the work defined by the project scope and captures all deliverables – internal, external, interim – in terms of the work to be completed, including project management.

Sample WBS:

1. Project Management

- 1.1. Work Breakdown Structure (WBS)
- 1.2. Roles & Responsibility Matrix
- 1.3. Change Control System

2. Reports / Documentation

- 2.1. Final Documentation Introduction
- 2.2. Literature / Markey Survey
- 2.3. Requirements Analysis
- 2.4. System Design
- 2.5. Implementation
- 2.6. Testing & Performance Evaluation
- 2.7. Conclusion & Outlook
- 2.8. End User Documentation
- 2.9. Application Administration Documentation
- 2.10. System Administrator Documentation

3. System

- 3.1. Development Environment
 - 3.1.1. IDE
 - 3.1.2. Version Control
 - 3.1.3. Server
 - 3.1.4. Database

3.2. Presentation Layer

3.2.1. Deliverable 1

3.2.2. Deliverable 2

3.2.3. ...

3.3. Business Logic Layer

3.3.1. Deliverable 1

3.3.2. Deliverable 2

3.3.3. ...

3.4. Data Management Layer

3.4.1. Deliverable 1

3.4.2. Deliverable 2

3.4.3. ...

3.5. Physical Layer

3.5.1. Deliverable 1

3.5.2. Deliverable 2

3.5.3. ...

Roles & Responsibility Matrix:

The purpose of roles & responsibility matrix is to identify who will do what.

WBS #	WBS Deliverable	Activity #	Duration (# of Days)	Responsible Team Member(s) & Role(s)
1.1	Requirement Gathering	Collect functional & non-functional requirements, finalize project scope and constraints	7	Farhan Ahmed (Team Lead, Analyst)
1.2	Roles & Responsibility Matrix	Define roles of team members, assign tasks, finalize management structure	3	All Three Members
2.1	Literature / System Survey	Research existing academic, program, university recommendation systems, identify research gaps and relevant datasets	7	Subhan Ahmed (Research Lead)
2.4	System Design	Design system architecture, methodology workflow, recommendation pipeline, UI mockups, and database/knowledge-base structure	6	Farhan Ahmed + Muhammad Zeeshan (System Designer)

3.1	Development Environment Setup	Setup IDE, GitHub, backend framework/server, database, and required AI/ML development environment	4	Muhammad Zeeshan (Backend/DB) + Subhan Ahmed (Version Control)
3.2	Presentation Layer	Develop user interface for student information, voice/text input, questions, recommendations, and university results	8	Farhan Ahmed (Frontend Developer)
3.3	Business Logic Layer	Implement student profile processing, NLP, speech-to-text, adaptive questioning, and preference extraction	8	Muhammad Zeeshan (NLP)
3.3	Business Logic Layer	Train and evaluate ML model for academic field recommendation	12	Subhan Ahmed (ML Developer)
3.3	Business Logic Layer	Implement LLM + RAG based academic program recommendation	10	Subhan Ahmed + Farhan Ahmed (LLM/RAG Developer)
3.3	Business Logic Layer	Implement university eligibility and preference-based filtering	6	Muhammad Zeeshan + Subhan Ahmed (Backend/Logic Developer)
3.3	Business Logic Layer	Implement MCDM/TOPSIS based personalized university ranking	8	All Members (MCDM/TOPSIS Developer)

3.4	Data Management Layer	Collect, preprocess, organize and maintain student, program and university datasets/knowledge base	7	Subhan Ahmed (Dataset) + Muhammad Zeeshan (DB Developer)
3.4	Data Management Layer	Develop university web-crawling and periodic data-update pipeline	7	Muhammad Zeeshan (Data/Web Crawling)
4.1	Testing & Development	Unit testing of student profile, NLP, recommendation modules, database and university filtering	5	Farhan Ahmed (Model Testing)
4.2	Testing & Development	Integration testing of frontend, backend, ML model, LLM/RAG, database and university ranking	7	All Three Members (Joint Responsibility)
4.3	Testing & Development	Performance testing including recommendation accuracy, response time, system usability and validation	5	Farhan Ahmed (System Testing) + Subhan Ahmed (Model Performance)
4.4	Final Deployment	Deploy final Edu Path system on localhost/cloud and prepare final demo & documentation	7	All Three Members (Joint Responsibility)

Approval

Project Supervisor

Comments _____

Name: **Mr. Nadeem Khan**

Date: **29th September, 2026**

Signature: _____

Project Coordinator

Comments _____

Name: **Mr. Mubariz Rehman**

Date: **29th September, 2026**

Signature: _____